

Notes: Chen, Fan, Gu, and Zhou (AER, 2020)
Arrival of Young Talent: The Send-Down Movement and Rural Education in China

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1 Big picture

- **Question.** Did the forced relocation of educated urban youth to rural China during the Cultural Revolution raise the educational attainment of rural children?
- **Historical shock.** The paper studies the *send-down movement*: about **16 million** urban youth were mandated to move to the countryside after 1968, and about **17.7 million** were sent down over 1962–1979.
- **Why this is economically interesting.** The program was not designed as an education policy. It was a political and labor-market response to urban chaos, unemployment, suspended schooling, and the need to move Red Guards out of cities. Yet the SDYs were much more educated than rural villagers and many became teachers or informal transmitters of urban knowledge and values.
- **Core empirical idea.** Counties received different densities of SDYs, and birth cohorts within a county differed in whether their school-age years overlapped with the movement. The paper combines these two sources of variation in a cohort difference-in-differences design.
- **Treatment.** County SDY intensity is

$$\%SDY_{c,p} = \frac{\text{number of SDYs received by county } c \text{ during 1968–1977}}{\text{county population in 1964}}.$$

The average SDY density is about **2.22%**.

- **Treated cohorts.** Rural children born in **1956–1969** are treated because their primary-school ages overlapped with the massive send-down period. Cohorts born in **1946–1955** form the baseline control group.
- **Main result.** A higher local density of SDYs significantly increased rural children’s schooling. The baseline coefficient is **3.237** years of schooling per unit of SDY density. At the mean density of 2.22%, this equals about **0.072 additional years of education per rural child**.
- **Aggregate magnitude.** About **245 million** rural children were potentially exposed. The estimated effect therefore implies about **17.6 million additional person-years of schooling** in rural China.
- **Education-margin result.** SDY exposure raised primary-school completion by about **0.98 percentage points** and junior-high completion by about **1.70 percentage points** at the mean SDY density. Per 100 SDYs, this corresponds to roughly **20.9 additional primary graduates** and **36.2 additional junior-high graduates**.
- **Heterogeneity.** The effect is larger for girls and for less-educated counties. This implies the program did not merely raise average schooling; it also reduced educational inequality by gender and region.
- **Mechanism.** SDYs increased the local supply of teachers, especially non-state-funded teachers. The effect is particularly strong for secondary-school teachers, consistent with the finding that the biggest educational gain is at the junior-high margin.
- **Long-run result.** The effect declined after SDYs returned to urban areas but did not disappear. Exposure also predicts more internal locus of control, higher probability of seeking education beyond junior high, higher-skilled occupations, more teachers among locals, later marriage, fewer children, and better education of the second generation.

- **Main takeaway.** A temporary inflow of educated people can have persistent human-capital effects in poor areas by relaxing teacher constraints, changing aspirations, and leaving behind institutions and norms.

2 Institutional background and data (key objects)

2.1 The send-down movement

- **Before the Cultural Revolution.** The idea of sending urban intellectuals to the countryside began in the 1950s and was initially smaller and more voluntary. From 1962 to 1966, about **1.29 million** urban youth were sent down.
- **After 1968.** The movement became massive and mandatory. Urban high-school graduates had limited schooling and job options because the Cultural Revolution disrupted schools, abolished college entrance exams, and created urban unemployment.
- **Scale.** From 1962 to 1979, around **17.7 million** SDYs were sent to rural areas, collective farms, or state farms. The main empirical analysis focuses on SDYs sent to rural villages during 1968–1977, about **11.5 million** people.
- **Temporary nature.** Most SDYs returned to cities after the movement ended in the late 1970s. This matters because it rules out the simplest mechanical interpretation that rural areas later looked more educated only because better-educated SDYs were counted as rural residents.
- **Destination assignment.** Most moves were within province; only **7.9%** of SDYs were sent outside their home province. Thus, county-level and intra-province variation are central.
- **Why education can be affected.** SDYs were junior-high or senior-high graduates in a rural environment where many adults had only primary schooling or less. Although they were officially sent to farm, many were assigned to technical, cultural, medical, or teaching jobs.

2.2 Local gazetteers: measuring county SDY exposure

The authors digitize information from over **3,000 local gazetteers**. These gazetteers are county-level historical volumes that record major local events and administrative statistics.

- The paper records the **total number of SDYs received by each county during 1968–1977**.
- The treatment variable divides this number by the county population in **1964**, the census year before the Cultural Revolution and the massive send-down movement.
- The core sample starts from 2,521 counties in the 1990 population census, excludes Beijing, Tianjin, Shanghai, and city-governed districts that are likely to be sending rather than receiving areas, and obtains SDY information for **1,843 of 2,039** remaining counties.
- After excluding counties without 1964 population data, the main county sample contains **1,773 counties**.
- Gazetteers also provide auxiliary variables: teachers, schools, grain production, and education expenditures. These are used for mechanisms and robustness tests.

Interpretation.

- The paper’s main data contribution is to move the send-down analysis from province-level flows to detailed county-level exposure.
- This is important because most SDY flows were within province; province-level data would miss most of the relevant geographic variation.

2.3 Population census data: measuring outcomes

The main education outcome comes from the **1 percent sample of the 1990 China Population Census**.

- The main sample is rural hukou individuals born between **1946 and 1969**.
- Treated cohorts: born **1956–1969**.
- Control cohorts: born **1946–1955**.
- The final rural sample has about **2.78 million** observations; the comparison urban sample has about **420,000** observations.
- Main outcome: imputed **years of education**, based on highest schooling level and completion status.
- Alternative outcomes: completion of primary school and completion of junior high school.

The 1990 census is useful because:

- by 1990 the treated cohorts had mostly completed schooling;
- most SDYs had already returned to cities, reducing mechanical composition concerns;
- migration was still limited before the mid-1990s, so current county is a reasonable proxy for schooling county.

2.4 Other datasets

- **1982 and 2000 censuses**. Used for pretrend and persistence checks. The 1982 census is close to the movement and useful for checking pre-existing cohort trends; the 2000 census is useful for cohorts born after 1970 and for marriage/fertility outcomes.
- **2010 census**. Used to study second-generation education.
- **2010 China Family Panel Studies (CFPS)**. Used to study values and locus of control.
- **Walder Cultural Revolution data**. Used to control for local severity of Cultural Revolution violence.

3 Empirical specifications (all equations + interpretation)

3.1 By-cohort DID / event-study specification

The first specification estimates a separate SDY effect for each birth cohort:

$$Y_{i,g,c,p}^{Edu} = \beta_0 + \sum_{\gamma=1946}^{1969} \beta_{1,\gamma} \%SDY_{c,p} \times \mathbf{1}(g = \gamma) + \beta_2 X_{i,g,c,p} + \lambda_c + \mu_{g,p} + \Lambda_c \times \mu_g + \varepsilon_{i,g,c,p}. \quad (1)$$

- $Y_{i,g,c,p}^{Edu}$ is education of individual i , born in cohort g , living in county c , province p .
- $\%SDY_{c,p}$ is county SDY density.
- $X_{i,g,c,p}$ includes individual controls, mainly gender and ethnicity.
- λ_c are county fixed effects.
- $\mu_{g,p}$ are province-by-cohort fixed effects.
- $\Lambda_c \times \mu_g$ interacts county baseline education with cohort dummies. Baseline education is measured using primary and junior-high graduation rates of the control group in each county.
- The omitted baseline cohorts are 1941–1945 in the event-study figure.

Interpretation.

- This is the paper’s graphical pretrend and dynamic-treatment specification.
- For cohorts born before 1956, $\beta_{1,\gamma}$ should be close to zero if counties with more SDYs were not already on different education trends.
- For cohorts born after 1956, the coefficients should rise if SDYs improved schooling during children’s primary-school years.

3.2 Main cohort DID specification

The main regression pools the affected cohorts into a single treatment group:

$$Y_{i,g,c,p}^{Edu} = \beta_0 + \beta_1 \%SDY_{c,p} \times \mathbf{1}(1956 \leq g \leq 1969) + \beta_2 X_{i,g,c,p} + \lambda_c + \mu_{g,p} + \Lambda_c \times \mu_g + \varepsilon_{i,g,c,p}. \quad (2)$$

Object of interest.

- β_1 compares treated versus control cohorts within the same county, and asks whether this cohort difference is larger in counties that received more SDYs.
- The identifying assumption is a **parallel-trend assumption**: absent the movement, education differences between treated and control cohorts would not be systematically related to county SDY density.

Why this design is stronger than a simple cross-sectional comparison.

- Counties that received more SDYs were not randomly chosen.
- The DID design does not require the level of SDY density to be random.
- It requires that unobserved determinants of education do not generate different *cohort trends* correlated with SDY density after controlling for county fixed effects, province-cohort fixed effects, and baseline-education-by-cohort controls.

3.3 Teacher-supply mechanism regression

To test whether SDYs raised the local supply of teachers, the authors use county-year teacher data from local gazetteers:

$$\%Teachers_{t,c,p} = \beta_0 + \beta_1 \%SDY_{c,p} \times \mathbf{1}(t \geq 1968) + \lambda_c + \mu_{t,p} + \varepsilon_{t,c,p}. \quad (3)$$

- $\%Teachers_{t,c,p}$ is the number of primary or secondary teachers in county c and year t , divided by 1964 county population.
- λ_c are county fixed effects.
- $\mu_{t,p}$ are province-year fixed effects or province-specific year controls.
- The coefficient tests whether high-SDY counties experienced a larger increase in teachers after 1968.

Interpretation.

- The mechanism is not merely that rural education was generally expanding.
- The paper asks whether counties with more SDYs saw larger post-1968 teacher increases.
- The stronger effect for non-state-funded teachers is especially consistent with SDYs being absorbed into local, informal, or commune-supported schools.

3.4 Values and locus-of-control regression

To test whether SDYs changed attitudes, the paper estimates:

$$\begin{aligned} LOC_{i,g,c,p} = & \beta_0 + \beta_1 \%SDY_{c,p} \times \mathbf{1}(1956 \leq g \leq 1969) + \beta_2 X_{i,g,c,p} \\ & + \lambda_c + \mu_{g,p} + \Lambda_c \times \mu_g + \varepsilon_{i,g,c,p}. \end{aligned} \quad (4)$$

- LOC is a locus-of-control index from the CFPS.
- A higher value means a more internal locus of control: more belief that outcomes depend on education, effort, talent, hard work, and intellect rather than family background, connections, luck, or wealth.

Interpretation.

- This specification asks whether exposure to SDYs changed values, not only years of schooling.
- The estimates remain similar after controlling for education, suggesting that value transmission is not mechanically just the result of more years of education.

3.5 Second-generation regression

To study intergenerational effects, the authors match children to rural household heads in the 2010 census and estimate:

$$Y_{i,g,c,p}^{Edu} = \beta_0 + \beta_1 \%SDY_{c,p} \times \mathbf{1}(1956 \leq P_{i,g,c,p} \leq 1969) + \beta_2 X_{i,g,c,p} + \lambda_c + \mu_{g,p} + \Lambda_c \times \mu_g + \varepsilon_{i,g,c,p}. \quad (5)$$

- $P_{i,g,c,p}$ is the parent's birth year.
- The treatment indicator is based on whether the *parent* was in the SDY-exposed cohort.
- The outcome is the child's education.

Interpretation.

- The coefficient captures whether SDY exposure had effects beyond the directly exposed generation.
- Potential channels include parental schooling, parental attitudes toward education, local teacher supply, marriage timing, and family size.

4 Identification and credibility (what makes the estimates believable)

4.1 The central identification problem

The main threat is that counties receiving more SDYs may have been different from other counties in ways that also affected educational trends.

- SDY placement was mandatory for urban youth, but not necessarily random across destinations.
- Destination choices reflected distance, local absorption capacity, labor needs, food availability, and sometimes military/frontier considerations.
- Therefore, the paper does *not* claim that SDY density is randomly assigned.

The key claim is narrower: after conditioning on rich fixed effects and baseline education trends, there is no evidence that counties with more SDYs had different cohort-specific education trends before treatment.

4.2 Why the cohort DID helps

- County fixed effects absorb permanent differences across counties: geography, historical education levels, local institutions, and long-run development.
- Province-cohort fixed effects absorb cohort shocks common within province: province-level education policies, cohort-specific economic conditions, and broad regional trends.
- Baseline-education-by-cohort controls allow counties with different initial education levels to follow different cohort trends.

- Event-study estimates show no meaningful relationship between SDY density and education for pre-treated cohorts.

4.3 Parallel-trend evidence

The event-study coefficients in Figure 3 are the most important credibility evidence.

- Before the 1956 cohort, coefficients fluctuate around zero.
- Starting with the 1956 cohort, coefficients rise as cohorts have more primary-school overlap with SDYs.
- Effects peak for cohorts most exposed during school age and decline for post-1970 cohorts after SDYs begin to leave.

Interpretation. The timing lines up with the movement. This is difficult to explain with a purely cross-sectional county difference.

4.4 Alternative explanations and how the paper addresses them

- **Grain productivity.** Counties with higher food production could receive more SDYs and also have better nutrition/education trends. The paper controls for 1965 grain production per capita interacted with treated cohorts; SDY estimates remain similar.
- **Local government education effort.** If pro-education counties accepted more SDYs, the estimates might reflect local government effort. The paper collects education expenditures and finds no evidence that SDYs increased local education spending. It also controls directly for school construction.
- **Rural school expansion.** Rural secondary schools expanded rapidly during the same period. The paper controls for local primary and secondary school construction rates. SDY effects remain strong. Moreover, SDYs and school construction appear to substitute: SDY effects are larger where secondary-school construction was weaker.
- **Cultural Revolution severity.** The Cultural Revolution disrupted schooling and was related to the movement. The authors control for local severity using the number of victims relative to 1964 population. The SDY coefficient is essentially unchanged.
- **Great Famine.** Famine exposure could affect cohort sizes and long-run human capital, and food productivity could be related to SDY placement. The paper controls for local famine severity based on cohort-size deficits. Results remain robust.
- **Migration.** The main analysis excludes people not living in their hukou registration county or prefecture. Since migration was still limited in 1990, this is less severe than it would be in later census years.
- **Mechanical composition.** About 5% of SDYs stayed in rural areas. This cannot mechanically explain the result because remaining SDYs were too few relative to the 245 million potentially affected rural children, and many SDYs belonged to older cohorts rather than the treated cohorts.
- **SDY-local marriages and offspring.** The overlap between SDYs' children and the treated cohorts is limited, and SDY marriage rates in rural areas were low during the relevant period.

4.5 Remaining caveats

- County-level SDY counts from gazetteers may contain measurement error. The paper addresses this with data-quality checks and robustness exercises, but exact annual SDY stocks are unavailable.
- The treatment is cumulative county exposure, not individual-level contact with SDYs. The estimates should be interpreted as reduced-form local exposure effects.
- The design estimates effects on rural children in receiving counties; it does not evaluate the costs to SDYs themselves.
- If the most educated rural people migrated away before 1990, the estimates may be conservative.

5 Mechanisms and interpretation

5.1 Teacher-supply channel

The clearest mechanism is that SDYs expanded the local teacher pool.

- SDYs were much more educated than rural residents.
- Rural China suffered from a shortage of teachers, especially during the Cultural Revolution when intellectuals were politically attacked.
- Many SDYs were assigned to cultural, technical, medical, or teaching jobs rather than ordinary agricultural work.
- Gazetteer data show that high-SDY counties experienced larger post-1968 increases in teachers.

Why the secondary-school effect is important.

- Primary-school access was already expanding before SDYs arrived.
- Junior-high access was much more limited.
- Therefore, extra educated youth mattered especially for the junior-high margin, where teachers were scarcer and the marginal return to teacher supply was higher.

5.2 Role-model and values channel

SDYs may have changed rural children's aspirations.

- Urban youth brought new information about education, cities, jobs, and family planning.
- Female SDYs were nearly as educated as male SDYs, which may have made them especially powerful role models for rural girls.
- The larger effects for girls and less-educated counties are consistent with a role-model or aspiration channel.
- CFPS evidence shows that exposed cohorts have more internal locus of control and stronger beliefs in the importance of education.

5.3 Persistence channels

The paper proposes three sources of persistence after SDYs returned to cities.

- **Infrastructure.** SDYs helped found or operate local schools. Even after they left, the school infrastructure and routines could remain.
- **Knowledge and values.** Information, attitudes toward schooling, and norms about work and family could persist after direct contact ended.
- **Indirect spillovers.** Directly exposed cohorts later became teachers and parents, transmitting education-oriented values to other rural residents and the next generation.

6 Tables and figures

6.1 Figure 1: Number of SDYs by resettlement, 1962–1979

Read: The figure shows the timing and destination of the send-down movement.

- The movement is small before the Cultural Revolution.
- It sharply increases around 1968–1969 after Mao’s directive.
- There is another major peak around 1974.
- Most SDYs went to rural villages, not only collective or state farms.
- The movement declines sharply after Mao’s death and after the central government begins ending the program.

Why it matters. The figure justifies the paper’s timing: 1968–1977 is the key exposure window, and primary-school-age overlap is a natural way to define treated cohorts.

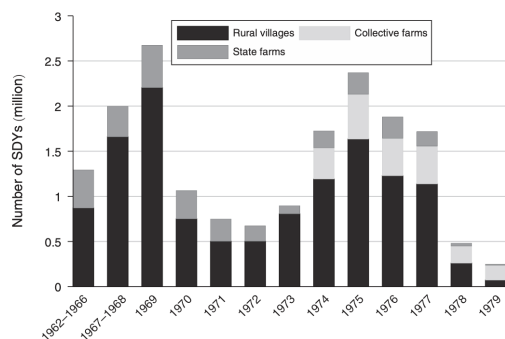


FIGURE 1. NUMBER OF SDYs BY RESETTLEMENT, 1962–1979

Source: Gu (2009)

TABLE 1—TOTAL NUMBER OF SENT AND RECEIVED SDYs IN EACH PROVINCE, 1962–1979

Province	SDY received (thousands)			SDY sent (thousands)		
	Total	Inside	Outside	Total	Inside	Outside
Beijing	384.2	384.2	0.0	636.3	384.2	252.1
Tianjin	193.6	193.6	0.0	465.1	193.6	271.5
Hebei	510.5	377.8	132.7	384.4	377.8	6.6
Shanxi	312.9	264.3	48.6	264.3	264.3	0.0
Inner Mongolia	299.3	193.8	105.5	193.8	193.8	0.0
Liaoning	2,018.0	2,013.4	4.6	2,013.4	2,013.4	0.0
Jilin	1,052.6	991.4	61.2	991.4	991.4	0.0
Heilongjiang	1,922.2	1,519.2	403.0	1,519.2	1,519.2	0.0
Shanghai	532.3	532.3	0.0	1,252.2	532.3	719.9
Jiangsu	861.2	810.2	51.0	828.4	810.2	18.2
Zhejiang	595.9	563.9	32.0	646.2	563.9	82.3
Anhui	725.5	576.5	149.0	576.5	576.5	0.0
Fujian	372.3	372.3	0.0	372.3	372.3	0.0
Jiangxi	622.5	504.5	118.0	504.5	504.5	0.0
Shandong	492.7	492.7	0.0	512.9	492.7	20.2
Henan	673.0	673.0	0.0	673.0	673.0	0.0
Hubei	878.6	878.6	0.0	886.6	878.6	8.0
Hunan	635.8	635.8	0.0	635.8	635.8	0.0
Guangdong	973.2	973.2	0.0	973.2	973.2	0.0
Guangxi	434.8	434.8	0.0	434.8	434.8	0.0
Sichuan	1,427.4	1,427.4	0.0	1,472.4	1,427.4	45.0
Guizhou	224.1	213.5	10.6	213.5	213.5	0.0
Yunnan	339.1	232.5	106.6	232.5	232.5	0.0
Tibet	3.4	3.4	0.0	3.4	3.4	0.0
Shaanxi	490.3	463.1	27.2	463.1	463.1	0.0
Gansu	264.3	245.2	19.1	245.2	245.2	0.0
Qinghai	51.0	43.6	7.4	43.6	43.6	0.0
Ningxia	57.5	49.2	8.3	49.2	49.2	0.0
Xinjiang	416.6	277.6	139.0	277.6	277.6	0.0
Total	17,764.8	16,341.0	1,423.8	17,764.8	16,341.0	1,423.8

Source: Gu (2009)

6.2 Table 1: Total number of sent and received SDYs by province

Read: Table 1 documents the scale and geography of the movement.

- Total SDYs: **17.7648 million**.
- Within-province SDYs: **16.3410 million**.
- Outside-province SDYs: **1.4238 million**.
- Therefore, only about **7.9%** were inter-province moves; **92.1%** were within-province moves.
- Shanghai, Beijing, and Tianjin sent many more SDYs than their population shares would suggest because they were highly urbanized.

Why it matters. The table motivates the paper’s county-level data construction. A province-level analysis would miss most of the movement because most flows occurred within provinces.

6.3 Figure 2: Number of received SDYs in each county

Read: Figure 2 maps cross-county variation in received SDYs.

- Some counties received very few SDYs; others received thousands or more than 10,000.
- The variation is geographically uneven and not simply a national uniform policy intensity.

Why it matters. This cross-county variation is one leg of the DID design. The other leg is differential cohort exposure within each county.

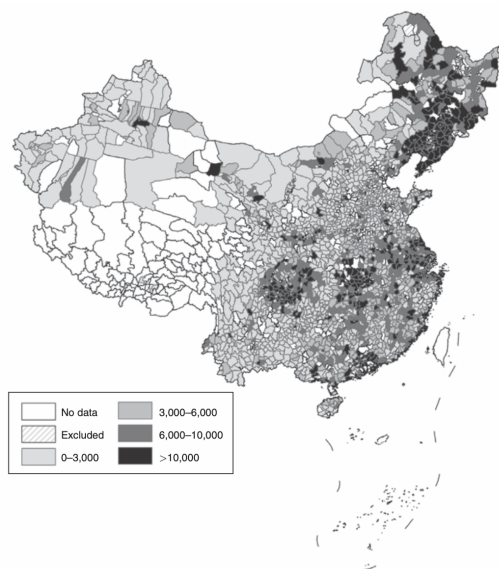


FIGURE 2. NUMBER OF RECEIVED SDYs IN EACH COUNTY

TABLE 2—SUMMARY STATISTICS OF THE 1 PERCENT SAMPLE FROM THE 1990 CENSUS

Cohort:	Control group (1946–1955)		Treatment group (1956–1969)	
	Rural (1)	Urban (2)	Rural (3)	Urban (4)
<i>Hukou:</i>				
Years of education	5.372 (3.365)	8.881 (3.178)	7.190 (3.100)	10.536 (2.511)
Complete primary school	0.617 (0.486)	0.911 (0.285)	0.799 (0.401)	0.972 (0.164)
Complete junior high	0.205 (0.404)	0.670 (0.470)	0.454 (0.498)	0.904 (0.294)
Male = 1	0.507 (0.500)	0.592 (0.492)	0.505 (0.500)	0.567 (0.495)
Han ethnic = 1	0.924 (0.266)	0.930 (0.255)	0.914 (0.280)	0.925 (0.264)
Age	39.031 (2.823)	39.208 (2.862)	26.638 (4.004)	26.993 (3.966)
Observations	960,127	148,244	1,815,752	271,603

Note: Standard deviations are in parentheses.

6.4 Table 2: Summary statistics from the 1990 census

Read: Table 2 establishes the rural-urban education gap and the main sample sizes.

- Rural control cohorts have **5.372** years of education on average; urban control cohorts have **8.881**.
- Rural treated cohorts have **7.190** years; urban treated cohorts have **10.536**.
- Rural primary completion rises from **0.617** in the control group to **0.799** in the treated group.
- Rural junior-high completion rises from **0.205** to **0.454**.
- The rural sample has **960,127** control observations and **1,815,752** treated observations.

Interpretation. Rural education improved dramatically across cohorts, so the empirical challenge is to separate the SDY contribution from broad cohort trends. This is why the paper uses county fixed effects, province-cohort fixed effects, and baseline-education trends.

6.5 Figure 3: Effect of SDYs on educational attainment by cohort

Read: Figure 3 is the event-study evidence.

- Pre-1956 coefficients are close to zero.
- The coefficients rise starting with the 1956 cohort, consistent with the first cohort whose primary schooling overlapped with SDYs.
- The effects are cumulative: cohorts with more overlap show larger effects.
- The effects decline for cohorts born after 1970, when SDYs gradually leave, but do not fall to zero.
- The same qualitative pattern appears in the 1982, 1990, and 2000 censuses.

Why it matters. Figure 3 is the most direct evidence for the timing-based identification story. It supports parallel trends and shows that the effect starts exactly when the treated cohorts should begin to be affected.

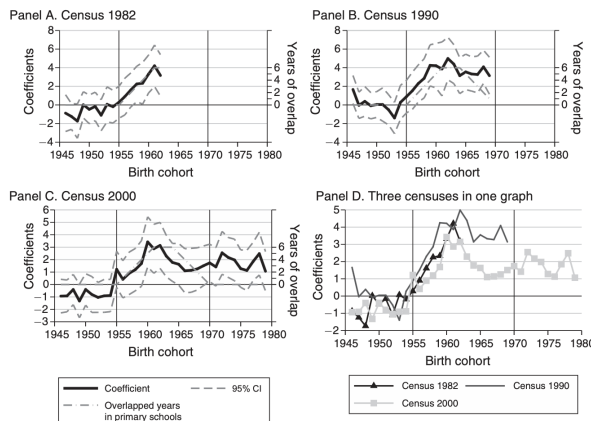


FIGURE 3. EFFECT OF SDYs ON THE EDUCATIONAL ATTAINMENT OF DIFFERENT COHORTS

Note: The left y-axis represents the coefficients from equation (1), which captures the effect of SDYs densities on different cohorts.

TABLE 3—THE EFFECT OF SDYs ON THE EDUCATIONAL ATTAINMENT OF RURAL CHILDREN (1990 CENSUS)

Dependent variables:	Years of education		Complete primary		Complete junior high		Placebo I (1990) (1946–1950) versus (1951–1955)	Placebo II (2000) (1970–1974) versus (1975–1979)
Sample:	Rural (1)	Urban (2)	Rural (3)	Urban (4)	Rural (5)	Urban (6)	(7)	(8)
Local density of received SDYs × affected cohorts (1956–1969)	3.237 (0.701)	0.151 (0.517)	0.441 (0.0873)	−0.0658 (0.0611)	0.767 (0.121)	−0.0517 (0.103)		
Local density of received SDYs × affected cohorts (placebo)							−0.817 (0.576)	−0.432 (0.319)
Male	1.874 (0.0284)	0.668 (0.0256)	0.201 (0.00361)	0.0319 (0.00227)	0.203 (0.00285)	0.0546 (0.00316)	2.286 (0.0300)	0.665 (0.0150)
Han ethnic	0.150 (0.0565)	3.34e-05 (0.0811)	0.0213 (0.00769)	0.00962 (0.00540)	0.00657 (0.00679)	0.0177 (0.00875)	0.0802 (0.0554)	0.477 (0.0401)
Observations	2,775,858	417,883	2,775,858	417,883	2,775,858	417,883	960,123	947,025
R ²	0.293	0.225	0.258	0.106	0.212	0.198	0.267	0.216
$\bar{Y}$ of control group	5.372	8.882	0.616	0.911	0.205	0.670		
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Province-cohort FE	✓	✓	✓	✓	✓	✓	✓	✓
Base education × cohort FE	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Standard errors are clustered at the county level. Local density of received SDYs is computed by dividing the number of received SDYs by the county population in 1964. Base education is calculated as the primary and junior high graduation rates of the control group.

6.6 Table 3: Main effect on rural education

Read: Table 3 estimates equation (2) for education outcomes.

- Years of education, rural sample: coefficient **3.237** with standard error **0.701**.
- Years of education, urban sample: coefficient **0.151** with standard error **0.517**; not significant.
- Primary completion, rural sample: coefficient **0.441**.
- Junior-high completion, rural sample: coefficient **0.767**.
- Urban coefficients for primary and junior-high completion are small and insignificant.
- Placebo tests using pre-treated cohorts and post-treated cohorts are insignificant.

Magnitude.

- At mean SDY density 2.22%, the years-of-schooling effect is

$$3.237 \times 0.0222 \approx 0.072.$$

- With 245 million potentially affected rural children, this implies

$$245 \times 0.072 \approx 17.6$$

million additional person-years of schooling.

- At the mean SDY density, primary completion increases by

$$0.441 \times 0.0222 \approx 0.0098,$$

about **0.98 percentage points**.

- Junior-high completion increases by

$$0.767 \times 0.0222 \approx 0.0170,$$

about **1.70 percentage points**.

Interpretation. SDYs affected rural children in destination counties but not urban residents in the same counties, which fits the historical mechanism.

6.7 Table 4: Heterogeneous effects

Read: Table 4 asks who benefited more.

- Male coefficient: **2.514**; female coefficient: **4.092**.
- Less-educated counties: **5.167**; more-educated counties: **2.059**.
- In less-educated counties, SDYs affect both primary and junior-high completion strongly: **0.838** and **0.882**.
- In more-educated counties, the primary effect is smaller: **0.153**; the junior-high effect remains larger: **0.579**.

Interpretation.

- The largest gains accrue to groups with the lowest initial education.
- This implies SDYs reduced educational inequality across gender and across county development levels.
- The stronger effect in counties with larger urban-rural education gaps supports the human-capital-spillover interpretation.

TABLE 4—HETEROGENEOUS EFFECT OF SDYs (1990 CENSUS)

Dependent variables:	Years of education				Primary		Junior	
	Male	Female	Less-educated counties	More-educated counties	Less-educated counties	More-educated counties	Less-educated counties	More-educated counties
Sample:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Local density of received SDYs $\times$ affected cohorts (1956–1969)	2.514 (0.624)	4.092 (1.007)	5.167 (1.594)	2.059 (0.668)	0.838 (0.184)	0.882 (0.217)	0.153 (0.0678)	0.579 (0.133)
Observations	1,403,169	1,372,686	1,349,489	1,426,369	1,349,489	1,349,489	1,426,369	1,426,369
R^2	0.197	0.316	0.294	0.194	0.253	0.192	0.141	0.188
Mean education years of control group	6.492	4.221	4.361	6.309	4.361	4.361	6.309	6.309
Primary graduate of control group	0.750	0.480	0.466	0.757	0.466	0.466	0.757	0.757
Junior high graduate of control group	0.302	0.105	0.140	0.265	0.140	0.140	0.265	0.265
Individual controls	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Province-cohort FE	✓	✓	✓	✓	✓	✓	✓	✓
Base education $\times$ cohort FE	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Only rural sample is used. Standard errors are clustered at the county level. Individual controls include gender and ethnicity. More-/less-educated counties are defined according to whether the average years of education of the control group exceed 5.5 years. Local density of received SDYs is computed by dividing the number of received SDYs by the county population in 1964. Base education is calculated as the primary and junior high graduation rates of the control group.

TABLE 5—EFFECTS OF SDYs ON THE SUPPLY OF LOCAL TEACHERS AND EDUCATIONAL FISCAL EXPENSES, 1955–1977 (LOCAL GAZETTEERS)

Dependent variables:	Density of primary school teachers			Density of secondary school teachers			Educational expenses
	All	State-funded	Non-state-funded	All	State-funded	Non-state-funded	
School types:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Local density of received SDYs $\times$ post 1968	0.034 (0.012)	0.005 (0.011)	0.051 (0.028)	0.017 (0.005)	0.010 (0.004)	0.035 (0.007)	0.072 (0.668)
Observations	6,521	1,823	1,799	6,336	1,261	1,205	11,677
R^2	0.809	0.754	0.839	0.811	0.776	0.725	0.855
Number of counties	488	152	151	493	107	104	769
County FE	✓	✓	✓	✓	✓	✓	✓
Province-year FE	✓	✓	✓	✓	✓	✓	✓

Notes: Standard errors are clustered at the county level. We collect both county-level information on the number of teachers and the number of SDYs from local gazetteers. Local density of received SDYs is computed from dividing the number of received SDYs by the county population in 1964. Local densities of primary and secondary school teachers are computed in a similar fashion.

6.8 Table 5: Teacher supply and education expenditures

Read: Table 5 tests the teacher channel using county-year gazetteer data.

- All primary teachers: coefficient **0.034**.
- All secondary teachers: coefficient **0.017**.
- Non-state-funded primary teachers: coefficient **0.051**.
- Non-state-funded secondary teachers: coefficient **0.035**.
- Education expenditures: coefficient **0.072** with standard error **0.668**; not significant.

Magnitude.

- At the teacher-sample mean SDY density, SDYs increased primary-school teachers by about **0.75 per thousand residents**.
- They increased secondary-school teachers by about **0.38 per thousand residents**.
- In 1965, there were only about **4.9 primary teachers** and **0.55 secondary teachers** per thousand residents.

Interpretation. The SDY effect works more through local teacher supply than through county fiscal spending. The secondary-teacher result matches the stronger effect on junior-high completion.

6.9 Table 6: Confounding factors

Read: Table 6 adds major historical and local confounders to the main DID.

- Controlling for 1965 grain production: SDY coefficient **3.779**.
- Controlling for primary and secondary school construction: coefficient **4.424**.
- Adding SDY-density-by-school-construction interactions: coefficient **4.513**.
- Controlling for Cultural Revolution severity: coefficient **3.196**.
- Controlling for Great Famine severity: coefficient **3.244**.
- Controlling for all major confounders simultaneously: coefficient **4.447**.

Interpretation.

- The SDY effect is not driven by grain productivity, school expansion, Cultural Revolution severity, or Great Famine exposure.
- School construction itself matters, especially secondary-school construction, but it does not absorb the SDY effect.
- The paper estimates that receiving about **43.2 SDYs** has the same effect as building one new secondary school. The 11.5 million SDYs sent to rural villages are equivalent to roughly **266,000** secondary schools in this calculation.

Dependent variables:	Years of education					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A</i>						
Local density of received SDYs × affected cohorts (1956–1969)	3.779 (0.876)	4.424 (1.096)	4.513 (1.182)	3.196 (0.725)	3.244 (0.701)	4.447 (1.342)
Grain production p.c. in 1965 × affected cohorts (1956–1969)	−0.0401 (0.0534)					0.0996 (0.123)
Local primary school construction rate ^a × affected cohorts (1956–1969)		−0.157 (0.217)	−0.133 (0.216)			−0.235 (0.242)
Local secondary school construction rate ^a × affected cohorts (1956–1969)		2.101 (1.069)	2.230 (1.069)			2.232 (1.198)
Local primary school construction rate × SDY density × affected cohorts (1956–1969) ^b			18.34 (9.470)			
Local secondary school construction rate × SDY density × affected cohorts (1956–1969) ^b			−77.74 (28.78)			
Local severity of Cultural Revolution × Cultural Revolution cohorts I (1954–1961) ^c				−0.462 (0.241)		−0.500 (0.617)
Local severity of Cultural Revolution × Cultural Revolution cohorts II (1962–1968) ^c				0.236 (0.632)		−0.169 (0.755)
Local severity of Great Famine ^d × Famine cohorts I (1955–1958)					−0.229 (0.0572)	−0.344 (0.0979)
Local severity of Great Famine ^d × Famine cohorts II (1959–1961)					−0.0553 (0.0663)	0.0635 (0.111)
Observations	1,964,222	1,243,049	1,243,049	2,759,926	2,775,804	978,074
R ²	0.297	0.301	0.301	0.294	0.293	0.304
<i>Panel B</i>						
Local density of received SDYs × affected cohorts (1956–1969)	3.710 (0.880)	4.617 (1.108)	4.617 (1.108)	3.216 (0.718)	3.237 (0.701)	4.887 (1.340)
Observations	1,964,222	1,243,049	1,243,049	2,759,926	2,775,804	978,074
R ²	0.297	0.301	0.301	0.294	0.293	0.304
Information on grain production	✓					✓
Information on number of schools		✓	✓			✓
Information on Cultural Revolution				✓		✓
Individual controls	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓
Province-cohort FE	✓	✓	✓	✓	✓	✓
Base education × cohort FE	✓	✓	✓	✓	✓	✓

Notes: Only rural sample is used. Standard errors are clustered at the county level. Individual controls include gender and ethnicity. Local density of received SDYs is computed by dividing the number of received SDYs by the county population in 1964. Base education is calculated as the primary and junior high graduation rates of the control group.

^a See Section IVC for the detailed construction of this variable.

^b To keep the double-interaction terms having a standard interpretation, the school construction rate and SDY density in the triple-interaction term use the deviation from the sample mean.

^c Local severity of the Cultural Revolution is proxied by dividing the number of victims by the county population in 1964. The data source is Walder (2017).

^d Local severity of the Great Famine is proxied by 1 minus the ratio of 1959–1961 cohort size over 1955–1957 cohort size.

Dependent variables:	Do you agree with following statements? 1 = strongly disagree, 5 = strongly agree									
	Internal locus of control					External locus of control				
	LOC index ^a	Education	Talent	Effort	Hard work	Intellect	Family SES	Family wealth	Family connection	Luck
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Panel A</i>										
Local density of received SDYs × affected cohorts (1956–1969)	13.186 (6.311)	2.817 (0.948)	0.728 (1.514)	0.716 (0.812)	0.791 (0.593)	−0.772 (0.688)	−2.092 (1.064)	−0.707 (2.021)	−1.438 (1.016)	−2.735 (1.166)
Observations	6,026	6,026	6,026	6,026	6,026	6,026	6,026	6,026	6,026	6,026
R ²	0.175	0.139	0.133	0.147	0.157	0.151	0.143	0.154	0.143	0.145
<i>Panel B</i>										
Local density of received SDYs × affected cohorts (1956–1969)	13.549 (6.188)	2.816 (0.946)	0.633 (1.481)	0.740 (0.809)	0.815 (0.602)	−0.767 (0.685)	−2.185 (1.028)	−0.851 (1.918)	−1.475 (0.998)	−2.818 (1.177)
Observations	6,026	6,026	6,026	6,026	6,026	6,026	6,026	6,026	6,026	6,026
R ²	0.185	0.139	0.143	0.149	0.158	0.151	0.152	0.173	0.144	0.152
Years of education	0.117 (0.013)	−0.000 (0.003)	−0.031 (0.004)	0.008 (0.003)	0.008 (0.003)	0.002 (0.003)	−0.030 (0.004)	−0.046 (0.004)	−0.012 (0.004)	−0.027 (0.005)
Individual controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Province-cohort FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Base education × cohort FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Only rural sample is used. Standard errors are clustered at the county level. Individual controls include gender and ethnicity. Local density of received SDYs is computed from dividing the number of received SDYs by the county population in 1964. Base education is calculated as the primary and junior high graduation rates of the control group.

^a Following Chen, Lu, and Xie (2018), we define the LOC index as the summation of ten z-scores (external locus-of-control statements take negative values). The higher the score, the more internal the individual. Online Appendix Table A2 presents the ten questions for generating this index.

TABLE 8—THE LASTING EFFECT OF SDYs ON OUTCOMES OTHER THAN EDUCATION

Dependent variables: Census:	>Junior high (conditional on junior high)	Labor force participation		High-skilled occupation	
	1990	1990		1990	
	(1)	(2)	(3)	(4)	(5)
Local density of received SDYs × affected cohorts (1956–1969)	0.481 (0.0830)	0.0409 (0.0390)	0.0365 (0.0391)	0.0622 (0.0167)	0.0355 (0.0168)
Years of education			0.00137 (0.000180)		0.00816 (0.000153)
Observations	1,147,340	2,775,858	2,775,858	2,633,659	2,633,659
R ²	0.071	0.182	0.182	0.012	0.040
$\bar{Y}$ of control group	0.174	0.951	0.951	0.0185	0.0185
Individual controls	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓
Province-cohort FE	✓	✓	✓	✓	✓
Base education × cohort FE	✓	✓	✓	✓	✓

Dependent variables: Census:	Teacher as occupation		Age at first marriage	Number of children ^a	Second generation's education ^b
	1990		2000		2010
	(6)	(7)	(8)	(9)	(10)
Local density of received SDYs × affected cohorts (1956–1969)	0.0398 (0.00993)	0.0250 (0.0103)	3.027 (0.698)	-0.957 (0.275)	4.186 (0.849)
Years of education		0.00456 (8.35e-05)			
Observations	2,775,858	2,775,858	2,219,527	1,007,793	108,164
R ²	0.005	0.024	0.172	0.186	0.547
$\bar{Y}$ of control group	0.00760	0.00760	22.73	2.770	9.012
Individual controls	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓
Province-cohort FE	✓	✓	✓	✓	✓
Base education × cohort FE	✓	✓	✓	✓	✓

Notes: Only rural sample is used. Standard errors are clustered at the county level. Individual controls include gender and ethnicity. Local density of received SDYs is computed by dividing the number of received SDYs by the county population in 1964. Base education is calculated as the primary and junior high graduation rates of the control group.

^a Fertility history information is only available to women aged 15–50.

^b See Section VF for the detailed construction of data that combine two generations.

6.10 Table 7: Locus of control and values

Read: Table 7 uses the 2010 CFPS to test whether exposure changed values.

- LOC index coefficient: **13.186**.
- Agreement that education raises future success: coefficient **2.817**.
- Belief in family socioeconomic status as destiny: coefficient **-2.092**.
- Belief in luck as destiny: coefficient **-2.735**.
- Results remain similar after controlling for own years of education.

Interpretation. SDYs did not only add schooling years. They also appear to have shifted rural residents' beliefs toward internal control, education, and effort.

6.11 Table 8: Lasting effects beyond basic education

Read: Table 8 studies long-run socioeconomic outcomes.

- Education beyond junior high, conditional on junior-high completion: **0.481**.
- Labor force participation: small and insignificant, consistent with already very high labor-force participation.
- High-skilled occupation: **0.0622**; after controlling for education, **0.0355**.
- Teacher occupation: **0.0398**; after controlling for education, **0.0250**.

- Age at first marriage: **3.027**.
- Number of children: **-0.957**.
- Second-generation education: **4.186**.

Interpretation.

- SDY exposure predicts outcomes that reflect both quantity and quality of human capital.
- Part of the occupation effect works through increased education, but not all of it disappears after controlling for schooling.
- The second-generation result suggests that the movement changed households and communities in ways that persisted after the directly exposed generation.

7 Short summary

- **Method.** Use county variation in SDY density and cohort variation in school-age exposure in a cohort DID design.
- **Main empirical result.** SDY exposure increases rural children’s schooling; at the mean SDY density, the effect is about **0.072 years** per child, or **17.6 million person-years** in aggregate.
- **Main educational margin.** The effect is especially important for junior-high completion, where rural teacher constraints were more binding.
- **Mechanism.** SDYs expanded the supply of local teachers, especially non-state-funded and secondary-school teachers.
- **Distributional result.** Effects are larger for girls and less-educated counties, so the program reduced educational inequality.
- **Persistence.** Effects decline after SDYs leave but remain positive, and exposed cohorts show changes in values, occupations, marriage, fertility, and children’s education.
- **Identification.** The main credibility evidence is the absence of pretrends in the event study and the robustness to controls for school expansion, grain production, Cultural Revolution severity, Great Famine severity, local education spending, migration, and mechanical composition.

8 Conclusion

Core conclusion. The send-down movement, despite being politically motivated and costly for many SDYs, unintentionally increased rural human capital. Educated urban youth raised schooling among rural children by serving as teachers, role models, and carriers of urban knowledge and values.

Economic intuition. In poor rural areas, the bottleneck was not only school buildings but also educated people who could teach and transmit aspirations. A temporary inflow of relatively educated youth relaxed that constraint. Because human capital is transmitted through schools, occupations, families, and values, even a temporary shock can have persistent effects.

Takeaway. The paper turns a major historical event into evidence on human-capital spillovers: moving educated people into low-education areas can raise schooling, reduce inequality, and generate effects that survive long after the original migrants leave. The analysis also warns that these benefits should not be confused with a full welfare evaluation of the movement, because the paper does not measure the costs imposed on the SDYs themselves.